Universidade do Algarve

Faculdade de Ciências e Tecnologia

Licenciatura em Engenharia Informática

Redes de Computadores I

Laboratório 5

Wireshark – TCP

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PL: 2

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Relatório

1. Capturing a bulk TCP transfer from your computer to a remote server

http://gaia.cs.umass.edu/wireshark-labs/alice.txt

http://gaia.cs.umass.edu/wireshark-labs/TCP-wireshark-file1.html

2. A first look at the captured trace

1. **What is the IP address and TCP port number used by the client computer (source) that is transferring the alice.txt file to gaia.cs.umass.edu? To answer this question, it's probably easiest to select an HTTP message and explore the details of the TCP packet used to carry this HTTP message, using the "details of the selected packet header window" (refer to Figure 2 in the "Getting Started with Wireshark" Lab if you're uncertain about the Wireshark windows).**

R: O cliente que transfere o ficheiro "alice.txt" tem endereço IP 10.20.85.2 e usa o porta TCP 59188.

2. **What is the IP address of gaia.cs.umass.edu? On what port number is it sending and receiving TCP segments for this connection?**

R: O servidor gaia.cs.umass.edu tem endereço IP 128.119.245.12 e usa o porta TCP 80.

3. TCP Basics

3. **What is the sequence number of the TCP SYN segment that is used to initiate the TCP connection between the client computer and gaia.cs.umass.edu? (Note: this is the "raw" sequence number carried in the TCP segment itself; it is NOT the packet # in the "No." column in the Wireshark window. Remember there is no such thing as a "packet number" in TCP or UDP; as you know, there are sequence numbers in TCP and that's what we're after here. Also note that this is not the relative sequence number with respect to the starting sequence number of this TCP session.). What is it in this TCP segment that identifies the segment as a SYN segment? Will the TCP receiver in this session be able to use Selective Acknowledgments (allowing TCP to function a bit more like a "selective repeat" receiver, see section 3.4.5 in the text)?**

R: O segmento SYN inicial tem raw sequence number 115689347. É identificado como SYN porque tem o bit SYN ativo nas flags TCP. O recetor pode usar Selective Acknowledgments, porque no SYN aparece a opção SACK Permitted.

4. **What is the sequence number of the SYNACK segment sent by gaia.cs.umass.edu to the client computer in reply to the SYN? What is it in the segment that identifies the segment as a SYNACK segment? What is the value of the Acknowledgement field in the SYNACK segment? How did gaia.cs.umass.edu determine that value?**

R: O segmento SYNACK tem raw sequence number 865652971. É identificado como SYNACK porque tem os bits SYN e ACK ativos. O campo Acknowledgment vale 115689348, que corresponde a 115689347 + 1, já que o SYN consome um número de sequência.

5. **What is the sequence number of the TCP segment containing the header of the HTTP POST command? Note that in order to find the POST message header, you'll need to dig into the packet content field at the bottom of the Wireshark window, looking for a segment with the ASCII text "POST" within its DATA field4,5. How many bytes of data are contained in the payload (data) field of this TCP segment? Did all of the data in the transferred file alice.txt fit into this single segment?**

R: O segmento que contém o cabeçalho do HTTP POST tem raw sequence number 115689348. Esse segmento transporta 14600 bytes de payload TCP. O ficheiro "alice.txt" não cabe nesse único segmento.

6. **Consider the TCP segment containing the HTTP "POST" as the first segment in the data transfer part of the TCP connection.**

- **At what time was the first segment (the one containing the HTTP POST) in the data-transfer part of the TCP connection sent?**
- **At what time was the ACK for this first data-containing segment received?**
- **What is the RTT for this first data-containing segment?**
- **What is the RTT value the second data-carrying TCP segment and its ACK?**
- **What is the EstimatedRTT value (see Section 3.5.3, in the text) after the ACK for the second data-carrying segment is received? Assume that in making this calculation after the received of the ACK for the second segment, that the initial value of EstimatedRTT is equal to the measured RTT for the first segment, and then is computed using the EstimatedRTT equation on page 242, and a value of □ = 0.125.**

Note: Wireshark has a nice feature that allows you to plot the RTT for each of the TCP segments sent. Select a TCP segment in the "listing of captured packets" window that is being sent from the client to the gaia.cs.umass.edu server. Then select: Statistics->TCP Stream Graph->Round Trip Time Graph.

R: O primeiro segmento de dados é enviado em 5.331560 s e o ACK correspondente chega em 5.442478 s, pelo que o RTT do primeiro segmento é 110.918 ms. O segundo segmento de dados é enviado em 5.442562 s e o ACK correspondente chega em 5.444882 s, pelo que o RTT do segundo segmento é 2.320 ms. O EstimatedRTT após o segundo ACK é 97.343 ms.

7. **What is the length (header plus payload) of each of the first four data-carrying TCP segments?**

R: 14620 bytes, 2940 bytes, 2940 bytes e 2940 bytes

8. **What is the minimum amount of available buffer space advertised to the client by gaia.cs.umass.edu among these first four data-carrying TCP segments7? Does the lack of receiver buffer space ever throttle the sender for these first four data-carrying segments?**

R: O menor receive window anunciado pelo recetor nos primeiros quatro segmentos de dados é 67072 bytes. Este valor não restringe o emissor, uma vez que a quantidade de dados em trânsito nesses primeiros envios é inferior ao espaço anunciado pelo recetor.

9. **Are there any retransmitted segments in the trace file? What did you check for (in the trace) in order to answer this question?**

R: Não se observam retransmissões TCP na captura. Os números de sequência evoluem de forma contínua e não há repetição de blocos de dados previamente enviados.

10. **How much data does the receiver typically acknowledge in an ACK among the first ten data-carrying segments sent from the client to gaia.cs.umass.edu? Can you identify cases where the receiver is ACKing every other received segment (see Table 3.2 in the text) among these first ten data-carrying segments?**

R: O recetor confirma, em geral, 1460 bytes por ACK. Isto verifica-se pelo aumento sucessivo do valor de acknowledgment em múltiplos de 1460 bytes, o que corresponde ao MSS negociado.

11. **What is the throughput (bytes transferred per unit time) for the TCP connection? Explain how you calculated this value.**

R: O throughput médio da ligação é aproximadamente 291874 bytes/s, o que corresponde a cerca de 291.9 kB/s ou 2.33 Mbit/s. Este valor resulta da razão entre o total de dados transferidos, 152983 bytes, e o intervalo de tempo entre o envio do primeiro segmento de dados e a receção do ACK que confirma o último byte enviado.

4. TCP congestion control in action

12. Use the Time-Sequence-Graph(Stevens) plotting tool to view the sequence number versus time plot of segments being sent from the client to the gaia.cs.umass.edu server. Consider the "fleets" of packets sent around t = 0.025, t = 0.053, t = 0.082 and t = 0.1. Comment on whether this looks as if TCP is in its slow start phase, congestion avoidance phase or some other phase. Figure 6 shows a slightly different view of this data.

R: Na captura de referência, os "fleets" observados em torno de t = 0.025, 0.053, 0.082 e 0.1 correspondem à fase de slow start. Esta conclusão resulta do crescimento rápido da quantidade de dados enviada em cada rajada sucessiva.

13. These "fleets" of segments appear to have some periodicity. What can you say about the period?

R: A periodicidade desses "fleets" é aproximadamente igual a um RTT. As rajadas são lançadas à medida que regressam os ACKs da rajada anterior, o que é consistente com o mecanismo normal de clocking do TCP.

14. Answer each of two questions above for the trace that you have gathered when you transferred a file from your computer to gaia.cs.umass.edu

R: Na captura analisada, o comportamento inicial também é compatível com a fase de slow start, embora a visualização seja influenciada pela agregação de segmentos. A periodicidade dominante dos "fleets" é de aproximadamente 0.11 s, valor que corresponde aproximadamente a um RTT da ligação.